

1. Constants

Problem Statement 1: `const` and `#define`

Write a C++ program to demonstrate the use of `const` and `#define` for defining constant values.

The program should:

1. Define the value of `PI` using `#define`.
2. Declare a constant integer `DAYS_IN_WEEK` using `const` and assign it the value `7`.
3. Display the values of `PI` and `DAYS_IN_WEEK`.
4. Declare a floating-point variable `radius` and assign it the value `5.0`.
5. Calculate the area of a circle using `PI` and `radius`.
6. Display the radius and calculated area.
7. Include a commented example showing that a `const` variable cannot be changed after initialization.

Expected Output:

```
Value of PI using #define: 3.14159
Days in a week using const: 7
Area of circle with radius 5 is: 78.5397
```

2. Type Casting

Problem Statement 1: Explicit Type Casting

Write a C++ program that:

1. Declares two integer variables `a = 5` and `b = 2`.
2. Declares a third variable `c` of type `float`.
3. Uses **explicit type casting** to convert one of the integer variables to `float` before performing the division.
4. Stores the division result in variable `c`.
5. Displays the value of `c`.

Expected Output

```
2.5
```